Bridging the Training Gap with Hyperion: A Training-Free Style Transfer Method on Diffusion Transformers

Yi Li, Chengyan Li, Zhirui Fang, Junchuan Zhang, Xuan Wang, Huaibo Huang, Yan Luo, and Yanqing Guo

Abstract—Training-free style transfer on Multimodal Diffusion Transformers (MM-DiT) remains challenging due to the persistent gap between trained adapters and injection-based training-free methods. Existing training-free methods typically perform forward feature injection, and thus struggle to reproduce the rich painterly textures achieved by trained adapters, often leading to only faint hue shifts. In this paper, we trace this limitation to the absence of backpropagation-induced spatial coupling and propose **Hyperion**, a fully training-free style-transfer pipeline for MM-DiT. At the optimization level, the pipeline introduces **Adam-in-loop**, which iteratively optimizes the diffusion latent at each denoising step with a frozen MM-DiT. This process recovers the gradient-driven coupling effect produced through the self-attention *value path*, where neighboring tokens receive highly similar updates and form coherent brushstroke textures. At the distribution level, the pipeline introduces **Sinkhorn Latent Transport (SLT)**, a single closed-form optimal-transport step that pre-aligns latent color distributions before Adam optimization, improving the operational efficiency of Adam-in-loop. To further enhance stylization quality, we develop a frequency-domain Sinkhorn loss to sharpen painterly textures and a layout-residual-suppression step to reduce layout interference. The pipeline can also be extended to image-to-image stylization. Extensive experiments on a 39-style $\times$ 6-prompt benchmark (234 pairs) demonstrate that Hyperion achieves CSD $= 0.640$ and CLIP-T $= 0.296$, improving CSD by **+5.2%** over Adam-8 at $2.5\times$ lower forward-pass cost and by **+16.6%** over Adam-only under an equal compute budget. Notably, Hyperion outperforms four pre-trained style adapters on **19 of 21** diverse style families, achieving a mean CSD of 0.657 versus 0.506 for the strongest baseline.

Index Terms—style transfer, diffusion transformers, optimal transport, latent optimization, training-free generation

1 Introduction

Style transfer on modern image generation models falls into two regimes that have so far remained largely separate. *Trained* methods (fine-tuned adapters, ControlNet-style encoders [1], [2], [3], [4]) inject a style reference into the model's weights or cross-attention banks via gradient descent, achieving high semantic fidelity and rich painterly texture. *Training-free* methods (attention K/V injection, token substitution, latent interpolation [5], [6], [7], [8], [9], [10], [11]) require no optimization and generalize immediately to any reference image, but their outputs exhibit only faint hue shifts with no discernible texture structure, and their style fidelity—measured by the Contrastive Style Descriptor (CSD) [12]—falls far below that of their trained counterparts. We argue that this gap originates from the underlying mechanism rather than from model scale or training data (Sec. 3.3).

We investigate *why* trained style methods produce coherent texture while training-free ones do not, and attribute the difference to the backpropagation graph. It is well established that self-attention keys and values in diffusion transformers encode both semantic content and fine-grained visual texture [13], [6], [14], [15]; manipulating K/V during denoising has therefore become the standard mechanism for lightweight style injection [5], [10]. During fine-tuning, however, gradients flow back through the full diffusion transformer and through its multi-head self-attention Jacobian [16]. Neighboring tokens carry nearly identical keys—a structural consequence of spatial continuity in the RoPE-encoded latent grid [17], [18]—so the rest of the image attends to them with nearly equal weights; and because each token's output is a weighted blend of value vectors, backpropagation through this *value path* therefore delivers nearly identical gradient signals to spatially adjacent tokens, causing their latent updates to point in the same direction. This spatial coherence is precisely what constitutes brushstroke texture: it mirrors the long-range spatial correlations that Gatys *et al.* [19] identify as the defining statistical signature of painterly style, now produced not by Gram-matrix optimization but as a natural by-product of the attention Jacobian during backpropagation. We make this claim precise in two ways: formally, by proving that spatially smooth content keys force adjacent tokens to receive near-identical gradient directions (Sec. 3.3, Appendix B); and empirically, by introducing the *spatial neighbor cosine*, a measure of how strongly adjacent per-token updates align, under which backward-pass gradients prove far more spatially coherent than any feedforward training-free update. Training-free methods, because they perform no backward pass, lack this cross-token coupling mechanism entirely.

- *Y. Li, C. Li, Z. Fang, J. Zhang, Y. Luo, and Y. Guo are with Dalian University of Technology, Dalian, China. E-mail: liyi@dlut.edu.cn, gggmdgzsx@mail.dlut.edu.cn, fangzr@mail.dlut.edu.cn, junchuanz@mail.dlut.edu.cn, 20211071090@mail.dlut.edu.cn, guoyq@dlut.edu.cn*
- *X. Wang is with Beihang University, Beijing, China. E-mail: zb2506519@buaa.edu.cn*
- *H. Huang is with the Institute of Automation, Chinese Academy of Sciences, Beijing, China. E-mail: huaibo.huang@cripac.ia.ac.cn*

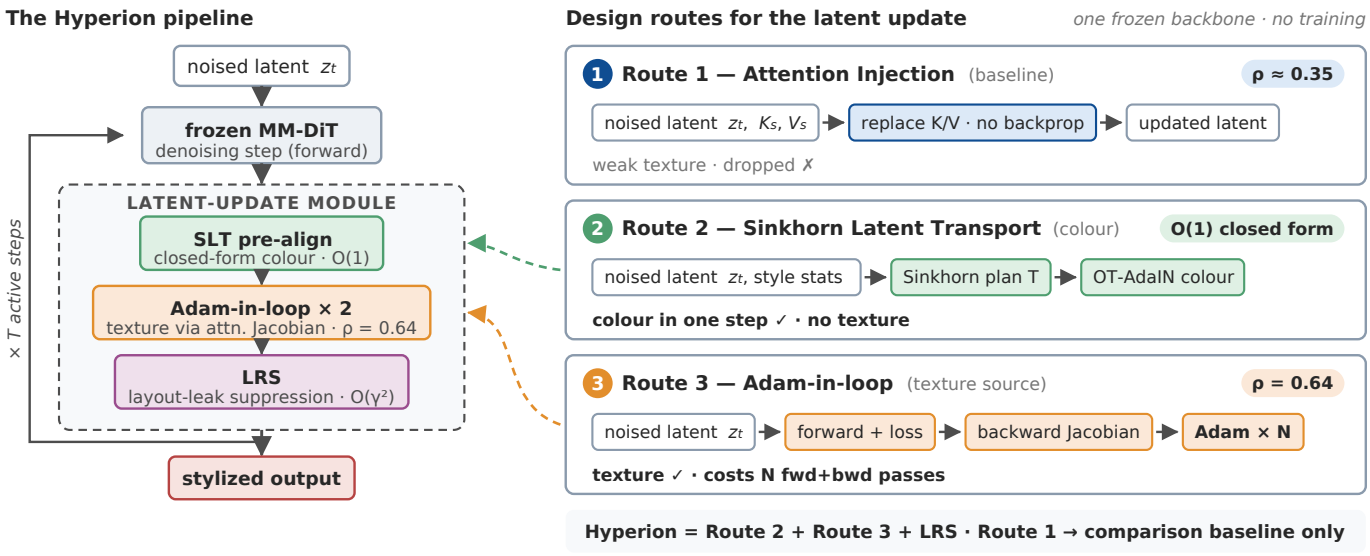


Fig. 1: Design space of the method. *Left* (dashed box = latent-update module): the final integrated **Hyperion pipeline**—at each active denoising step the frozen MM-DiT runs a style forward pass, SLT pre-aligns color in one closed-form step, two Adam steps inject cross-token texture through the self-attention Jacobian (ρ=0.64), and LRS suppresses layout leakage. *Right*: the three latent-update design routes, each shown as a detailed module framework—**Route 1** Attention Injection (the training-free baseline that replaces content K/V with style K/V in one block, no backprop; $\rho\approx$0.35, weak texture); **Route 2** Sinkhorn Latent Transport (closed-form color alignment, $O(1)$, no texture); and **Route 3** Adam-in-loop (backprop through the frozen backbone, the source of spatial coherence, ρ=0.64). The Hyperion integrates Route 2 and Route 3 (with LRS layout suppression), discarding the injection baseline.

This observation suggests a direct remedy: *introducing backpropagation into the training-free setting*. Starting from a standard inversion-interpolation baseline that blends content and style attention features without any optimization [5], [6], [10], [20], we add an iterative Adam update [21] loop at each denoising timestep. At each step, a style forward pass extracts reference K/V tensors; the content latent is then updated by Adam on a style-alignment loss computed from the content forward pass, with gradients flowing through the self-attention Jacobian [16]. We call this **Adam-in-loop** style optimization. To our knowledge, this is the first training-free pipeline on MM-DiT whose outputs exhibit brushstroke texture, raising style fidelity from the injection regime to a level comparable with trained adapters.

More iterations, however, mean slower generation: each Adam step costs a full forward and backward pass through the model, and accumulated style pressure gradually erodes faithfulness to the content prompt [22]. Profiling the optimization reveals that the early iterations are largely devoted to *color distribution alignment*: the gradient is dominated by a global low-frequency component that corrects bulk color statistics, leaving only a small fraction of each step for the intended texture signal. Recent work confirms that high- and low-frequency components of diffusion latents behave fundamentally differently during generation [23], [24], corroborating that a single gradient step cannot serve both goals efficiently—a frequency separation we later exploit directly through a frequency-domain Sinkhorn loss (Sec. 3.5).

This analysis reveals a mismatch between tool and task: gradient descent is well suited to optimizing a differentiable objective through a neural Jacobian, but poorly suited to global distribution alignment. Color mismatch between the content and style latents is not a local, high-curvature objective—it is a bulk statistical discrepancy between two token populations [25]. Optimal transport solves exactly this problem in closed form: it computes the globally cost-minimizing assignment of content tokens to style tokens that equates the two marginal distributions [26], [27], [28]. Crucially, optimal transport does not pass through the attention Jacobian at all—it operates directly in latent token space, bypassing the mechanism that generates spatial coherence. This property is precisely what is needed: optimal transport resolves, in a single step, the sub-task that consumes most of Adam's gradient budget, while leaving the Jacobian untouched for texture formation. We implement this as **Sinkhorn Latent Transport (SLT)**, a soft transport step driven by a doubly-stochastic Sinkhorn plan [29] whose marginal constraints guarantee that every style token contributes to the result, preventing the mode collapse that row-only softmax normalization permits [27]. SLT matches color distributions at negligible cost; yet, used alone, its outputs remain visually flat. The reason is structural: the transport update at each token is independent of every other token's update, so no mechanism couples adjacent positions, and the spatially coherent structures that constitute brushstroke texture cannot emerge (Sec. 3.6).

The two mechanisms are complementary: SLT resolves what Adam expends iterations on (global color distribution via optimal transport [26], [28]), while Adam produces what SLT structurally cannot (cross-token gradient coupling via the self-attention Jacobian [16]). Our **Hyperion** pipeline (Fig. 1) chains them within each denoising step: SLT pre-aligns the

color distribution, after which the remaining Adam iterations are directed almost entirely toward texture. We formalize this as a pre-alignment efficiency result (Sec. 3.8, Appendix C) and confirm it empirically: a Hyperion schedule with only two Adam iterations matches the style fidelity of a far longer Adam-only schedule at a fraction of the compute, and clearly outperforms Adam-only optimization at an equal compute budget.

Contributions:

- We identify the disparity between training-free and trained style transfer as a *backpropagation gap*: trained methods derive spatial coherence from the self-attention Jacobian, whereas training-free injection methods lack this coupling entirely. We support this diagnosis with a formal coherence analysis and a reusable diagnostic, the *spatial neighbor cosine* ρ, which we further extend to an output-image texture-coherence score (Sec. 4.3).
- We present **Hyperion**, a fully training-free style-transfer pipeline that bridges this gap by combining **Adam-in-loop** and **Sinkhorn Latent Transport (SLT)**. Adam-in-loop performs iterative backpropagation through a frozen MM-DiT at each denoising step, while SLT provides a closed-form optimal-transport color pre-alignment whose complementarity with Adam-in-loop is formally characterized. Frequency-domain Sinkhorn boosting and layout-residual suppression further enhance texture formation and suppress layout leakage.
- We validate Hyperion extensively on a 234-pair benchmark, where it raises CSD from < 0.15 (injection) to 0.640, exceeds an 8-iteration Adam-only schedule at $2.5\times$ lower forward-pass cost, outperforms four trained adapters on 19 of 21 style families, and extends effectively to image-to-image stylization. Controlled ablations further attribute the gains to the individual components.

2 Related Work

Style transfer methods divide into two regimes—training-free *injection* and *trained* adapters—separated by a persistent quality gap: injection transfers color but little texture, while trained adapters achieve rich texture at the cost of per-style optimization. We trace this gap to a single missing ingredient, gradient feedback through the self-attention value path (Sec. 3.3), and close it with a training-free pipeline combining Adam-in-loop optimization, Sinkhorn latent transport, and layout residual suppression. We organize the survey around this thesis, noting where each prior family sits relative to it.

2.1 Training-Free Style Transfer: from Gram Matrices to Attention Injection

Classical neural style transfer established the statistical view of style that we build on. Gatys *et al.* [19] showed that Gram matrices of VGG [30] features capture perceptual texture statistics; feed-forward networks [31] amortized the per-style cost, and AdaIN [25] reduced arbitrary style transfer to channel-wise moment matching, later extended to whitening–coloring transforms [32] and higher-order moments [33]. These CNN-based methods operate on fixed VGG hierarchies and cannot express the cross-token gradient coupling that we identify as the source of brushstroke texture in modern diffusion transformers [34], [35]—in particular MM-DiT [18], which processes image and text in jointly attending streams under RoPE positional encoding [17] and whose attention features encode high-fidelity spatial semantics [14], [15].

Because self-attention keys and values in diffusion models [36] encode both semantic content and fine-grained texture [13], manipulating K/V during denoising is the standard training-free stylization pathway. InST [7] encodes the style image in prompt space via textual inversion; Prompt-to-Prompt [13] established attention maps as the central control knob for semantic manipulation. StyleID [6] retains content queries while substituting style K/V with attention temperature scaling; StyleAligned [5] shares K/V across a batch; AttenST [37] aligns feature statistics with a content-aware AdaIN inside each attention block; SCSA [38] routes style tokens to content positions by semantic affinity; StyleSSP [10] manipulates the sampling start-point via frequency-domain inversion; and Stable Flow [20] restricts injection to structurally "vital" layers. V-Shuffle [8] spatially permutes value tokens to sever layout correspondences; Inversion-Free Dual RF [9] runs two rectified-flow branches with cross-branch injection; U-StyDiT [39] adapts MM-DiT with a multi-view style modulator. Mask- and energy-based extensions [40], [41], [42] trade user input or per-image overhead for finer control. All of these share one structural limitation: the latent update is a single feedforward substitution with no iterative feedback. Our spatial neighbor cosine analysis (Sec. 3.3) provides a unified explanation—without backpropagation through the self-attention Jacobian, no cross-token gradient coupling exists, and stylization remains confined to faint hue shifts.

2.2 Trained Style Adapters and Content–Style Disentanglement

A parallel direction trains lightweight adapter modules on style-paired data. IP-Adapter [1] projects CLIP image embeddings of the reference into a parallel cross-attention branch; InstantStyle [43] routes adapter features to style-correlated layers only; CSGO [4] trains explicit content- and style-branch heads on a curated paired dataset; StyleShot [44] replaces CLIP with DINO-derived [45] style features. ControlNet [2] and T2I-Adapter [46] attach trainable encoders for spatial conditioning; on flow-matching backbones, MegaStyle [3] and OmniStyle [47] scale trained stylization with large curated datasets, and per-style LoRA fine-tuning [48] remains the fine-tuned alternative. Trained methods achieve high fidelity when the target style overlaps their training data, but the data and GPU costs preclude rapid adaptation to unseen references; our method is complementary—zero training, zero external data, deployable on any MM-DiT checkpoint.

Within both regimes, a central challenge is separating a style's *texture statistics* from its *spatial layout*. SCAdapter [49] decouples the two via a CLIP-based separation module; DICE [50] learns a contrastive subspace decomposition of artist style; UniCSG [51] performs staged semantic and frequency disentanglement; SCSA [38] disentangles by semantic routing. Our **Layout Residual Suppression** (Sec. 3.7) addresses the same problem with a minimal closed-form operation—attenuating the position-specific key residual

$R_K^{(\ell)} = K_s - \mu_K$ by a layer-dependent coefficient while preserving the permutation-invariant token mean μ_K (global texture)—conceptually related to the mean–residual statistics of HSI [33] but operating directly on attention keys with no trainable parameters.

2.3 Latent Optimization at Inference Time

A minority of methods optimize the diffusion latent directly at inference time without updating model weights. SDEdit [52] adds noise to a reference image and runs the reverse process for image-guided synthesis. DNO [53] performs direct noise optimization with a probability-preserving regularizer, demonstrating that gradient-based latent updates can align generation with arbitrary reward signals at inference time; DAS [54] frames test-time alignment as sequential Monte Carlo sampling, avoiding reward over-optimization through principled importance weighting; FreeControl [11] extracts structural attention maps in a single forward pass and uses them to condition generation without iterative optimization. Our **Adam-in-loop** differs from all of the above: it optimizes the latent specifically via the *style-alignment loss gradient through the self-attention Jacobian*, thereby injecting spatial coherence as a structural property of the optimization path rather than as a reward or structural condition. To our knowledge, no prior work has identified the attention Jacobian as the mechanistic source of brushstroke texture or exploited it explicitly for training-free style transfer.

2.4 Optimal Transport in Generative Modeling

The Wasserstein distance [26] and entropic-regularized OT [29] are widely used in generative modeling as distribution discrepancy measures and alignment tools [27]. Wasserstein Flow Matching [28] extends flow-matching to families of distributions along Wasserstein geodesics, and rectified flow [18] minimizes transport cost in the coupling space of the reverse process—in both cases OT enters as a *training* objective. Our **SLT** is qualitatively distinct: it applies Sinkhorn optimal transport as a closed-form *per-step latent pre-alignment* within a fully frozen network, exploiting the doubly-stochastic constraint to equate content and style token distributions in a single pass—no training objective, no network gradients—removing the color-correction sub-task that consumes the bulk of Adam's early gradient budget and leaving subsequent iterations free to generate texture through the Jacobian.

3 Method

3.1 Setup and Notation

We target training-free style transfer on Klein 9B (FLUX.2 [klein] 9B [55], an open-weights MM-DiT rectified-flow model [18] with 8 double-stream blocks and 24 single-stream blocks). Given a text prompt p and a style reference image x_s, the goal is to generate an image that matches the semantic content of p while adopting the visual style of x_s, without modifying any model weights.

Notation: Let $z_t \in \mathbb{R}^{L\times D}$ denote the packed image latent at denoising timestep t, L tokens of dimension D. $z_s = \mathrm{VAE}(x_s)$ is the style-image encoding. Self-attention at layer ℓ produces $Q^{(\ell)}, K^{(\ell)}, V^{(\ell)} \in \mathbb{R}^{H\times L\times d}$ (H heads, $d = D/H$); subscripts c and s denote content and style branches. We run 25 denoising steps with guidance scale 3.5 at 512×512 on a single H200 GPU, fixed seed 43.

3.2 Attention Interpolation Baseline

The canonical training-free approach blends content and style attention features via a scalar interpolation:

$$\hat{z}_t = (1-\beta)\,\mathrm{Attn}(Q_c, K_c, V_c) + \beta\,\mathrm{Attn}(Q_c, K_s, V_s), \qquad \beta \in [0,1]. \tag{1}$$

This requires no optimization (one additional style forward pass per step) and represents the broad class of injection-based training-free methods [5], [8], [9].

Limitation. The latent update is a single feedforward operation. There is no iterative feedback to reinforce the style signal or propagate it across the spatial token grid. On our benchmark, best-effort injection achieves CSD < 0.15, giving only faint hue shifts with no texture structure. We trace this failure directly to the absence of backpropagation, as detailed in the next section.

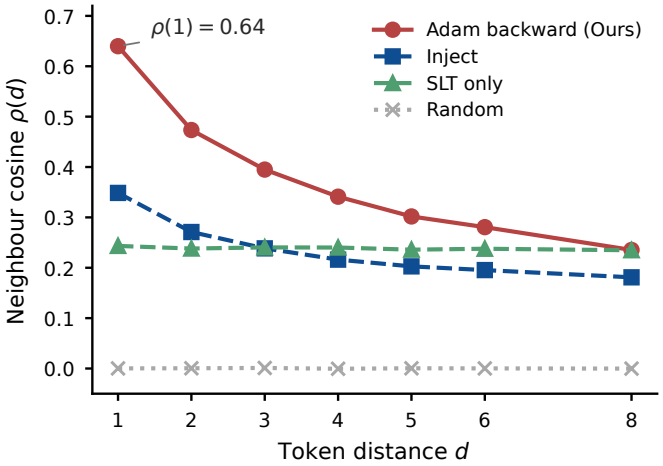
3.3 Spatial Coherence of Latent Updates

Trained style methods produce coherent brushstroke texture, with self-reported style fidelity of CSD $\approx$ 0.6–0.8 under their own protocols; under our unified protocol the strongest baseline reaches only a mean of 0.506 (Table 4). To understand why, consider the gradient that flows back through the self-attention module during fine-tuning. Backpropagation couples tokens through the *value path* of attention: token i's value contributes to the output at every position j with weight A_{ji}, so the gradient arriving at token i aggregates residuals from *all* positions, weighted by the i-th column of the attention matrix. Neighboring tokens $(i, i{+}1)$ carry nearly identical keys—RoPE encodes adjacent grid positions with nearly identical rotation matrices [17] ($\|\mathbf{R}(\mathbf{p}_i) - \mathbf{R}(\mathbf{p}_{i+1})\|_F \approx 0.098$) and denoising latents are spatially smooth—so adjacent attention columns nearly coincide and adjacent tokens receive nearly identical gradient signals. These correlated updates cause neighboring tokens to be nudged in the *same direction*, producing the long-range spatial correlations that define painterly texture.

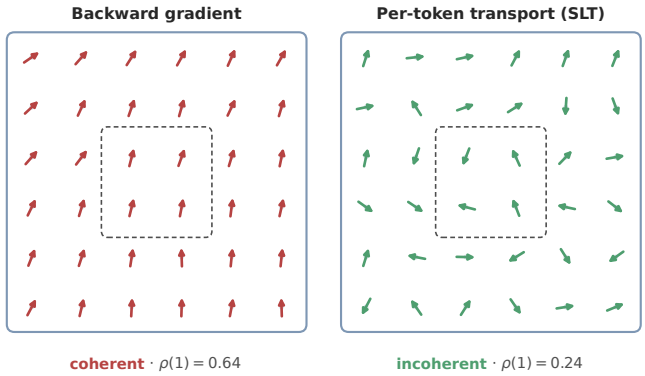
Empirical confirmation: We measure this directly via the *spatial neighbor cosine*:

$$\rho = \frac{1}{|\mathcal{N}|}\sum_{(u,v)\in\mathcal{N}} \cos(\delta_u, \delta_v), \tag{2}$$

where δ_u is the per-token latent update at grid position u and $\mathcal{N}$ is the set of horizontally and vertically adjacent position pairs on the 32×32 token grid ($|\mathcal{N}| = 2 \times 31 \times 32$). Results are shown in Table 1 and Fig. 2a, with the convergence and gradient-source analyses in Fig. 3; the detailed per-token update-direction coherence maps appear in the supplement (Fig. 15). Fig. 3a shows that Backward+SLT (ours) converges fastest, reaching at $N{=}2$ the style loss that backward-only optimization requires $N{\approx}4$ to match, while the gradient-free



(a) Measured spatial neighbour cosine $\rho(d)$ vs. token distance. Adam-backward updates are strongly coherent for neighbours (ρ=0.64) and stay coherent over distance; injection (0.35) is weaker, per-token SLT is flat-low (0.24, no local structure), and a random control is zero.



(b) Mechanism: the value-path self-attention Jacobian couples neighbouring tokens—adjacent content keys are nearly identical, so backpropagation delivers near-identical gradient directions to spatially adjacent positions, whereas per-token transport (SLT) has no such coupling (ρ=0.24; Table 1).

Fig. 2: Spatial coherence of latent updates: **(a)** the *measured* coherence $\rho(d)$ and **(b)** the value-path Jacobian *mechanism* that produces it. Detailed per-token coherence maps are in the supplement (Fig. 15).

Inject and SLT-only baselines cannot iterate. Fig. 3b isolates the source of the coherence: shuffling the spatial order of the style K/V leaves ρ unchanged (0.654 vs. 0.654), ruling out the style-image layout; removing style K/V entirely collapses coherence to ρ=0.168, whereas a single-patch style loss recovers most of it (ρ=0.628)—confirming that the self-attention backward pass propagates the style signal globally through the value path.

TABLE 1: Spatial neighbor cosine ρ (d=1) by gradient source, measured at the first Adam iteration and averaged over 6 content–style pairs (4 pairs for the content-loss-only and stop-grad-Q conditions). Fig. 3b visualizes the gradient-source conditions on the 4-pair ablation set (first iteration); minor value differences from this table reflect the smaller sample. Higher = more spatially coherent = brushstroke texture.

Gradient condition	ρ	Interpretation
Random direction (control)	0.00	No structure
SLT per-token OT	0.24	Accidental proximity
Content-loss-only gradient	0.17	Weak self-coupling
Adam full gradient (backprop)	0.64	**Value-path Jacobian**
Stop-grad-Q gradient	0.65	Value-path dominant

The 2.6× gap between Adam gradients ($\rho = 0.64$) and OT updates ($\rho = 0.24$) is the quantitative signature of the backpropagation gap: training-free OT operates token by token independently, whereas backward-pass gradients couple neighbors through the shared attention pattern and are therefore 2.6× more spatially coherent.

Proposition 3.1 (Spatial gradient coherence). *Let* $g_i = \partial \mathcal{L}_{\text{style}}/\partial z_i$ *be the gradient of the style-alignment loss at token* i. *The content self-attention produces a* value-path coupling term

$$g_i^{\text{val}} = -2W_V \sum_j A^c_{ji} R_j, \tag{3}$$

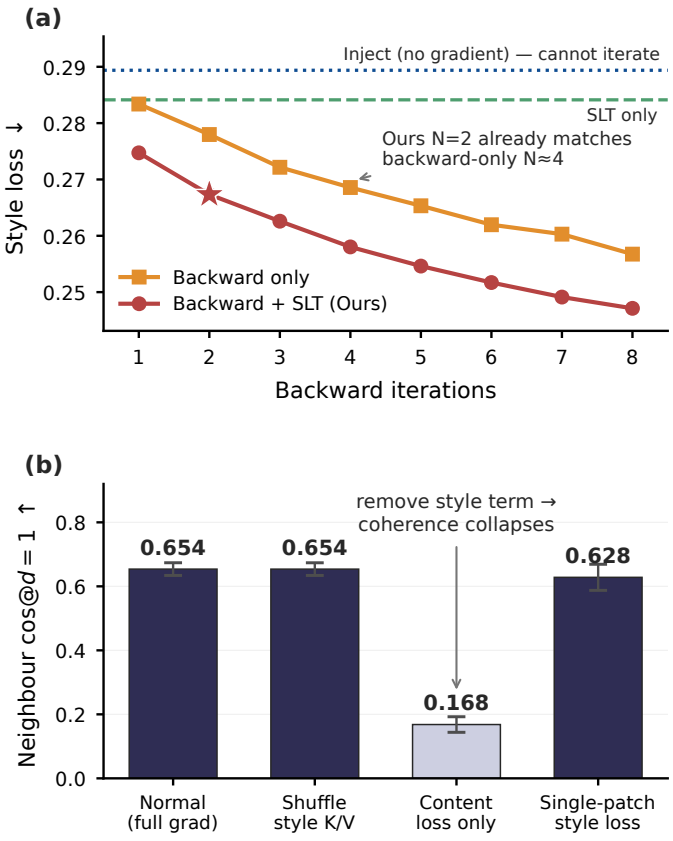


Fig. 3: Style-loss convergence (a) and the gradient-source analysis of spatial coherence (b); see Sec. 3.3.

where A^c *is the content attention matrix and* $R = O_s - O_c$. *If content keys satisfy the* ϵ*-spatial smoothness condition* $\|K_{c,i} - K_{c,i+1}\|_2 \le \epsilon\sqrt{d}$ *(with* $q_{\max}\epsilon \le 1$, $q_{\max}$ *the maximal query*

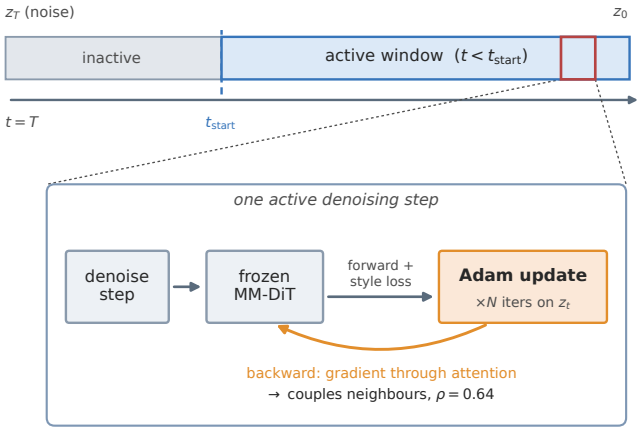


Fig. 4: The Adam-in-loop schedule: N Adam iterations inside each active denoising step ($t < t_{\text{start}}$).

norm), then

$$\cos\big(g_i^{\text{val}},\, g_{i+1}^{\text{val}}\big) \;\geq\; 1 - \frac{C\,\epsilon^2 \|R\|_F^2}{\|g^{\text{val}}\|_{\min}^2}, \qquad C = 2e^2 \|W_V\|_2^2\, q_{\max}^2\, L. \tag{4}$$

SLT updates have $\partial\Delta_i/\partial z_j = 0$ for $i \neq j$ (with the transport plan T held fixed during the update, as in our implementation), so no analogous coupling exists; their spatial coherence depends only on accidental proximity of transport targets.

The key step (Appendix B) is that g_i^{val} is a weighted average of the residual R over *all* positions, with column weights A_{ji}^c that are nearly equal for adjacent i whenever content keys are spatially smooth. This smoothness is structurally motivated in MM-DiT [18]: the RoPE rotation-difference bound [17] $\|R(\mathbf{p}_i) - R(\mathbf{p}_{i+1})\|_F \approx 0.098$ constrains the positional part of the key difference $\|K_{c,i} - K_{c,i+1}\|$, and the spatial smoothness of denoising latents constrains the remainder; the resulting attention-weight difference is bounded by the softmax logit-difference argument in Appendix B. SLT operates entirely in token-distribution space and bypasses the attention Jacobian; $\partial\Delta_i/\partial z_j = 0$ for $i \neq j$ by construction, which accounts for the $2.6\times$ gap in ρ.

Ablation — which gradient path drives spatial coherence: To isolate the value-path contribution experimentally, we compare three gradient conditions: (i) *Normal* (full gradient), (ii) *Stop-grad-Q* (Q detached before the attention product, removing the query-side gradient path), and (iii) *Random direction* (control). Normal and Stop-grad-Q achieve identical spatial coherence ($\rho \approx 0.65$) at every iteration, while Random yields $\rho \approx 0.00$ (see Supplementary Fig. 14). The query-side Jacobian path is therefore *not* the source of spatial coherence; combined with the empirical subdominance of the key-side term ($\|g^K\|/\|g^{\text{val}}\| < 0.15$, Appendix B), this identifies the value-path term g_i^{val} (Eq. 3) as the responsible mechanism, consistent with Proposition 3.1.

3.4 Adam-in-Loop Latent Optimization

The diagnosis suggests a direct fix: run backpropagation through the frozen model at each denoising step, optimizing only the latent z_t. No weights change; the gradient serves purely to propagate spatial coherence into the latent update. We call this **Adam-in-loop**. Fig. 4 illustrates the schedule: at every active step, gradients backpropagated through the frozen MM-DiT update the content latent, supplying the cross-token spatial coupling.

Formulation: At each denoising timestep $t < t_{\text{start}}$, after the standard denoising update, we run N Adam steps on a style-alignment loss:

$$\mathcal{L}_{\text{style}} = \sum_{\ell} \big\| \text{Attn}(Q_c^{(\ell)}, K_s^{(\ell)}, V_s^{(\ell)}) - \text{Attn}(Q_c^{(\ell)}, K_c^{(\ell)}, V_c^{(\ell)}) \big\|_1, \tag{5}$$

with an ℓ_1 content-anchor term to resist structural drift ($\mathcal{L} = w_s \mathcal{L}_{\text{style}} + w_c \mathcal{L}_{\text{cnt}}$, w_s=2.0, w_c=1.5, lr = 0.03). Gradients flow through the content forward pass and through the self-attention Jacobian, providing the spatial coherence that injection cannot.

Iteration-count and equal-budget comparisons against Adam-only—and the selection rule for the conservative Adam-8 reference configuration used in Tables 3 and 8—are analyzed in Sec. 4.5 (Table 7, Fig. 9).

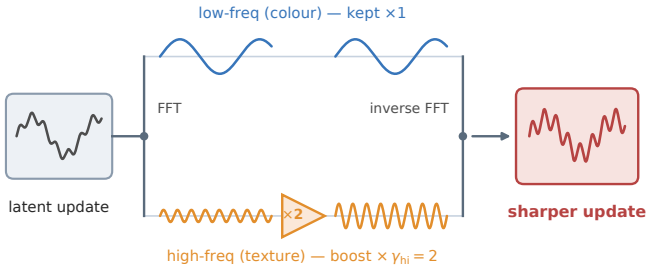


Fig. 5: The frequency-domain Sinkhorn loss: the high-frequency band is amplified (γ_{hi}=2) before transport.

3.5 Frequency-Domain Sinkhorn Loss

Before turning to Adam's efficiency bottleneck (Sec. 3.6), we describe one refinement internal to the style loss itself. Fig. 5 sketches the mechanism: amplifying the high-frequency (texture) band before transport sharpens brushstroke detail without disturbing the global color alignment. Within Adam's style loss $\mathcal{L}_{\text{style}}$, we decompose the latent into low- and high-frequency components via per-channel frequency separation and apply boosted transport weighting to the high-frequency component:

$$\mathcal{L}_{\text{style}} = \mathcal{L}_{\text{lo}} + \gamma_{\text{hi}}\, \mathcal{L}_{\text{hi}}, \quad \gamma_{\text{hi}} = 2.0. \tag{6}$$

The low-frequency component captures global tone; the high-frequency component carries fine-grained texture: thin brushstrokes, sharp grain, edge details. Standard ℓ_1 loss, averaged over all frequency bands, under-weights high-frequency terms because they carry less energy; the boost coefficient $\gamma_{\text{hi}} = 2.0$ corrects this, recovering fine-grained detail that would otherwise be smoothed by the gradient averaging in Adam. Ablation (Sec. 5) shows $+0.4\%$ CSD versus no boost.

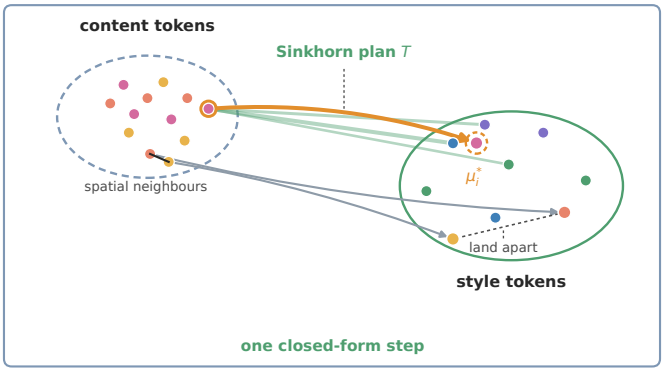


Fig. 6: Sinkhorn Latent Transport in the latent colour plane: the doubly-stochastic plan T (green fan, line width $\propto T_{ij}$) matches each content token to the whole style distribution and moves it to its OT-weighted target μ_i^* in one closed-form AdaIN step (≈0.1 ms, no forward pass, no attention Jacobian). Adjacent tokens (solid link) are transported independently and land apart: colour is aligned but no cross-token coupling arises ($\rho(1)$=0.24), hence no texture.

3.6 Sinkhorn Latent Transport

Gradient budget of early Adam iterations: Profiling the Adam-in-loop gradient at each iteration reveals that the first 2–3 steps are dominated by the correction of low-frequency color mismatch between z_t and z_s. Color distribution alignment is a global, per-channel operation that is inefficient to perform via spatially local per-token gradient steps. If this phase could be replaced by a single closed-form operation, subsequent Adam iterations could begin immediately in the texture regime.

SLT formulation: Optimal transport provides exactly this: the doubly-stochastic transport plan $T = \text{Sinkhorn}(\tilde{S}/\tau)$ computed from the cosine-similarity matrix $\tilde{S}_{ij} = \cos(\tilde{z}_t[i], \tilde{z}_s[j])$ over high-frequency residuals $\tilde{z} = z - \bar{z}$ (spatial mean subtracted) globally matches each content token to a distribution over style tokens in a single operation. We use T to derive per-token style distribution targets and apply adaptive instance normalization [25]:

$$\hat{z}_t[i] = \sigma_i^* \frac{z_t[i] - \mu_t}{\sigma_t} + \mu_i^*, \quad z_t \leftarrow (1-\alpha)\, z_t + \alpha\, \hat{z}_t, \qquad (7)$$

where $\mu_i^* = \sum_j T_{ij}\, z_s[j]$ and $\sigma_i^* = \sqrt{\sum_j T_{ij}\, z_s[j]^2 - (\mu_i^*)^2}$ are the OT-weighted per-token target mean and standard deviation, and μ_t, σ_t are the global statistics of the content latent (α=0.10, τ=0.02, n_{iters}=10). Compared to directly interpolating token embeddings ($z_t \leftarrow (1-\alpha)z_t + \alpha T z_s$), this OT-AdaIN formulation transfers per-token distribution statistics rather than embedding values, preserving content structure while ensuring global color alignment. The doubly-stochastic constraint ($\sum_j T_{ij}$=1, $\sum_i T_{ij}$=1) ensures every style token contributes to the transported statistics, preventing the mode collapse that row-only softmax normalization permits. The Sinkhorn solve runs in ≈ 0.1 ms with no model forward pass; the style features z_s are already available from the attention injection branch, so SLT adds zero additional forward passes. Fig. 6 depicts the transport: it operates directly in token space, bypassing the attention Jacobian—resolving color cheaply while, by construction, producing no cross-token spatial coupling.

Structural limitation: By construction the OT assignment is per-token independent: nothing couples the update of token i to its neighbor's state, so SLT updates carry low spatial coherence ($\rho = 0.24$, 2.6× below Adam gradients; Table 1) and no spatially coherent structures emerge. SLT resolves the global distribution gap but cannot generate *local structural correlations*; its measured quality ceiling and wall-clock cost are quantified in the ablation (Sec. 5) and the efficiency analysis (Sec. 4.5).

3.7 Layout Residual Suppression

Style K/V tensors carry both texture statistics and the spatial layout of the reference image. Injecting them unmodified causes content objects to drift toward reference subject positions (*layout leakage*).

Decomposition: Any style key tensor decomposes as $K_s^{(\ell)} = \mu_K^{(\ell)} + R_K^{(\ell)}$, where $\mu_K^{(\ell)}$ is the token mean (permutation-invariant, carries global style statistics) and $R_K^{(\ell)} = K_s - \mu_K$ is the position-specific residual (carries spatial layout).

LRS suppression: We attenuate the layout residual with a depth-dependent coefficient:

$$K_s^{(\ell)\prime} = \mu_K^{(\ell)} + \gamma^{(\ell)}\,(K_s^{(\ell)} - \mu_K^{(\ell)}), \qquad (8)$$

with the same transform applied to V_s; $\gamma^{(\ell)}$ is linearly interpolated over block depth from $\gamma_{\text{shallow}} = 0.7$ to $\gamma_{\text{deep}} = 0.5$. Deeper blocks, whose features are more layout-specific, are attenuated more strongly; the retained residual preserves texture directionality cues while suppressing the layout leakage that causes subject-position drift.

Why this suppresses leakage: Layout leakage arises because a content query q_i attends to the style keys with *position-dependent* weights, importing *where* the reference's structures sit. Decompose the pre-softmax logit under Eq. (8):

$$q_i^\top K_{s,j}^{(\ell)\prime} = \underbrace{q_i^\top \mu_K^{(\ell)}}_{\text{constant in } j} + \gamma^{(\ell)}\, q_i^\top R_{K,j}^{(\ell)}. \qquad (9)$$

The first term is identical for every style token j: it shifts all logits equally and therefore *cancels* in the softmax. Only the second, position-varying term selects which style positions a token attends to, and its dispersion across positions is scaled by γ^2:

$$\text{Var}_j\big(q_i^\top K_{s,j}^{(\ell)\prime}\big) = \big(\gamma^{(\ell)}\big)^2\, \text{Var}_j\big(q_i^\top R_{K,j}^{(\ell)}\big). \qquad (10)$$

As $\gamma^{(\ell)} \to 0$ the attention distribution over style tokens becomes uniform: the output still draws on the global style statistics carried by $\mu_K^{(\ell)}$ (palette, texture energy) but no longer on the *spatial arrangement* encoded in $R_K^{(\ell)}$, so the layout residual cannot propagate into the generated image. As $\gamma^{(\ell)} \to 1$ the full layout returns. LRS therefore continuously trades layout leakage against texture retention, and the depth schedule attenuates most strongly in the deep, layout-specific blocks while preserving the shallow-block residual that carries brushstroke directionality. Appendix I makes the leakage bound precise.

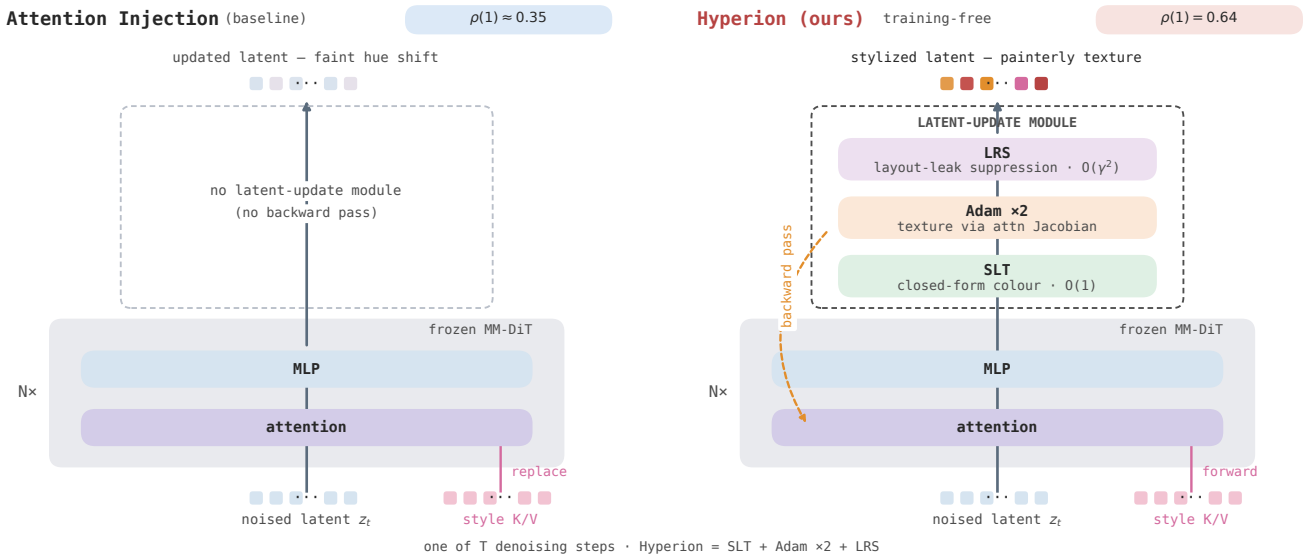


Fig. 7: One denoising step of the attention-injection baseline (left) and Hyperion (right); the two routes differ only in the latent-update module above the frozen backbone.

3.8 Hyperion Pipeline: SLT Pre-Alignment with Adam Refinement

Design: The two mechanisms are complementary with non-overlapping failure modes: SLT is efficient at what Adam is slow at (global color alignment), while Adam produces what SLT cannot (spatial texture coherence via the Jacobian). The Hyperion pipeline (Fig. 1) chains them within each denoising step in three stages. Fig. 7 draws one such step next to the injection baseline: the two routes differ only in whether the latent-update module fills the lane above the frozen backbone. Both panels share the same skeleton—the noised latent z_t and the style K/V tokens enter the frozen MM-DiT—but injection applies no latent update and yields only a faint hue shift ($\rho(1)\approx 0.35$), whereas the chain SLT$\to$Adam$\times 2\to$LRS produces spatially coherent texture ($\rho(1)=0.64$). First, the style branch supplies $K_s^{(\ell)}, V_s^{(\ell)}$ and LRS layout-leakage suppression is applied (Sec. 3.7). Second, the SLT pre-alignment (Eq. 7, $\alpha = 0.10$) collapses the color distribution gap before Adam begins. Third, two Adam steps are run on $\mathcal{L} = 2.0\,\mathcal{L}_{\text{style}} + 1.5\,\mathcal{L}_{\text{cnt}}$ with frequency-domain Sinkhorn loss (Sec. 3.5); because the color gap is already closed, both iterations are spent entirely on texture, generating brushstrokes through the Jacobian. For completeness, we also disclose two auxiliary color operations that bookend this loop, both applied unconditionally for Hyperion: a per-step AdaIN warm-start toward the (noised) style latent statistics, applied right after LRS and before SLT at every active step ($\alpha_w = 0.5$); and a terminal one-dimensional color correction applied once, after the denoising loop, that matches the final latent's global per-channel statistics to those of the (clean) style latent (Algorithm 1). Removing both costs only $0.683 \to 0.647$ CSD (-0.036) on the 40-sample ablation subset, while CLIP-T is essentially unaffected (in fact rises slightly, $0.274 \to 0.283$); the primary quality gain is therefore attributable to SLT and Adam-in-loop rather than to these color-correction auxiliaries.

Proposition 3.2 (Pre-alignment iteration efficiency). *Decompose the style loss as $\mathcal{L} = \mathcal{L}_c + \mathcal{L}_t$ (color and texture components), with near-orthogonal gradients $\langle g_c, g_t\rangle \approx 0$. Assume (A1) $\|g_c(z^{(0)})\| \geq \kappa \|g(z^{(0)})\|$ with $\kappa \approx 0.85$, (A2) $\|g_c(z^{\text{SLT}})\| \leq \epsilon \|g(z^{\text{SLT}})\|$ with $\epsilon \ll 1$, and (A3) the SLT displacement changes g_t by at most a relative $\delta \ll 1$. Model the optimizer as normalized gradient steps $z^+ = z - \eta\, g/\|g\|$ (the abstraction matching Adam's magnitude-invariant step size). Then, to first order in η, the per-step texture-loss decrease satisfies*

$$\Delta_t^H \geq \eta\,(1 - 2\epsilon - 2\delta)\,\|g_t\| \quad \textit{(Hyperion, at } z^{\text{SLT}}\textit{)}, \tag{11}$$

$$\Delta_t^A \leq \eta\,\sqrt{1-\kappa^2}\,\|g_t\| \quad \textit{(pure Adam, at } z^{(0)}\textit{)}, \tag{12}$$

so Hyperion's guaranteed texture progress per iteration exceeds the pure-Adam ceiling by at least $(1 - 2\epsilon - 2\delta)/\sqrt{1-\kappa^2} \approx 1.9\times$ in the idealized regime $\kappa = 0.85$, $\epsilon, \delta \ll 1$.

The intuition: a normalized optimizer spends a fixed step length per iteration, and at $z^{(0)}$ the dominant color gradient consumes most of that budget—the component of the step aligned with texture is at most $\sqrt{1-\kappa^2} \approx 0.53$. After SLT, $g_c \approx 0$, so essentially the entire step serves texture. This $\approx 1.9\times$ per-step advantage is an idealized estimate: at the directly profiled constants ($\kappa \approx 0.67$, $\delta \approx 1.6$, gradient cosine ≈ 0.26; Appendix C, Remark C.1) the lower bound of Eq. (11) is no longer positive, so the ratio should be read as a mechanism statement about where the step budget goes rather than as a pointwise numeric guarantee—but the empirical effect is unambiguous: 2 Hyperion iterations outperform 2 pure Adam iterations by $+16.6\%$ CSD on the ablation subset. The complete derivation is in Appendix C.

The complete pipeline is given in Algorithm 1; headline comparisons on the full benchmark and the equal-budget analysis appear in Secs. 4.2 and 4.5.

3.9 Extension to Image-to-Image Style Transfer

The T2I Hyperion pipeline extends to image-conditioned (I2I) inputs with three modifications; all other components (SLT, Adam, LRS, freq-boost) are unchanged. (i) the content

Algorithm 1 Hyperion Pipeline

Require: prompt p, style image x_s, frozen backbone Φ
$z_s \leftarrow \mathrm{VAE}(x_s)$; sample $z_T \sim \mathcal{N}(0, I)$
for $t = T, T-1, \ldots, 1$ **do**
  $z_t \leftarrow \mathrm{denoise}(\Phi(z_t, t, p))$
  if $t < t_{\text{start}}$ **then** ▷ $t_{\text{start}} = 850$ for T2I
    $\{K_s^{(\ell)}, V_s^{(\ell)}\} \leftarrow \Phi\text{-fwd}(z_s^{(t)})$
    Apply LRS: $K_s^{(\ell)} \leftarrow \mu_K + \gamma^{(\ell)}(K_s^{(\ell)} - \mu_K)$ ▷ $\gamma^{(\ell)}$: $0.7 \to 0.5$ by depth
    AdaIN warm-start: $z_t \leftarrow (1-\alpha_w)\, z_t + \alpha_w\, \mathrm{AdaIN}(z_t,\, z_s^{(t)})$ ▷ α_w=0.5, toward style stats
    $T \leftarrow \mathrm{Sinkhorn}(\cos(\tilde{z}_t, \tilde{z}_s)/\tau)$, $\tilde{z}=z-\bar{z}$ ▷ τ=0.02, 10 iters
    $z_t \leftarrow \text{OT-AdaIN}(z_t, z_s;\, T,\, \alpha{=}0.10)$ ▷ Eq. 7
    for $n = 1, 2$ **do**
      forward $\Phi(z_t) \to Q_c$, compute $\mathcal{L}$ ▷ freq-domain Sinkhorn loss
      $z_t \leftarrow \mathrm{AdamStep}(z_t,\, \nabla_{z_t}\mathcal{L})$ ▷ lr = 0.03
    end for
  end if
end for
$z_0 \leftarrow \mathrm{AdaIN}_{1D}(z_0,\, z_s)$ ▷ terminal color correction, gated on for Hyperion
return $\mathrm{VAE}^{-1}(z_0)$

image conditions the denoising process via in-context token concatenation rather than latent initialization; (ii) the per-step AdaIN warm-start of Algorithm 1 is reused unchanged in form, with α_w lowered from 0.5 to 0.3; and (iii) the injection start time is shifted from $t_{\text{start}} = 850$ to 870.

Content conditioning via in-context tokens: As in T2I, the initial latent z_T is pure Gaussian noise with no structural prior; the content image instead enters as in-context conditioning rather than as a latent initialization. We encode $z_c = \mathrm{VAE}(x_c)$ once and, at every denoising step, concatenate it to the denoising latent along the token axis before the forward pass, assigning it distinct positional ids from the image latent's own; only the L-token prefix of the model's output is kept for the update:

$$\hat{v}_t = \Phi\big(\mathrm{concat}(z_t,\, z_c),\, t,\, p\big)[:L], \tag{13}$$

where L is the token length of z_t and $[:L]$ truncates the model's per-token prediction to the image-latent prefix, discarding the prediction at the content-token positions. Crucially, z_c itself is never denoised and never mixed into the statistics of z_t: the content image's structure reaches the output only through the attention that z_t's tokens pay to the z_c tokens at every forward pass, not through any modification of z_t's initial distribution or per-channel statistics. This concatenation runs at every denoising step unconditionally, independent of t_{start}.

Re-weighted style warm-start and start time: The per-step AdaIN warm-start toward the (noised) style latent statistics introduced in Algorithm 1 is reused unchanged in form for I2I; only its strength is lowered, $\alpha_w : 0.5 \to 0.3$, since with content structure already supplied at every step by the in-context tokens above, a weaker color pre-set suffices before SLT begins. The T2I start time $t_{\text{start}} = 850$ is calibrated for noise-initialized latents, where 25 steps of denoising from pure noise are needed before semantic structure is coherent enough for style injection. For I2I, structural content is already available to every denoising step through the concatenated tokens described above—rather than being encoded in the initial latent—so meaningful content features are present one step earlier; we set $t_{\text{start}} = 870$ (one additional step) to exploit this, giving the content branch a one-step head-start against the style signal. All other hyperparameters (α=0.10, τ=0.02, N=2, γ_{hi}=2.0, $\gamma^{(\ell)}$=0.7→0.5) are unchanged.

4 Experiments

4.1 Experimental Setup

Model and hardware: All experiments use Klein 9B (MM-DiT architecture, 8 double-stream + 24 single-stream blocks, 32 layers total) via the DiffSynth backend, run in bf16 mixed precision on a single H200 GPU. Outputs are generated at 512×512 resolution with 25 denoising steps, guidance scale 3.5, and fixed seed 43. No model parameters are modified.

Benchmark construction: We assemble a benchmark of 40 style references (`sty/00–sty/39`) curated to span the breadth of visual styles a training-free method must handle: traditional media (oil impasto, watercolor, ink line art, bronze relief, ukiyo-e), graphic and illustrative styles (flat-color posters, screenprint, art-deco, constructivist propaganda, comic), digital and pixel art, and photographic textures. References were chosen for mutual visual distinctness so that style fidelity is not inflated by near-duplicate references. We exclude `sty04` from all T2I evaluation because its near-uniform pixel statistics make the CSD embedding degenerate, leaving 39 styles. The six T2I content prompts (Table 2) are fixed in advance to span object, animal, still-life, landscape, and architectural categories, decoupling measured style fidelity from any single content type. The full style bank is shown in the supplement (Fig. 12).

Evaluation splits: *T2I* pairs every retained style with all six prompts ($39 \times 6 = 234$ pairs). *Ablations and sweeps* use a 40-sample subset—8 styles (`sty` 00/10/14/20/27/28/33/38) $\times$ 5 prompts (`kp0–kp4`)—chosen to span the style categories above while permitting rapid iteration; the multi-seed robustness study reuses this subset. *I2I* uses a 40-pair subset: the same 8 styles $\times$ 5 content images (`cnt` 00/04/08/12/17).

TABLE 2: T2I benchmark content prompts.

ID	Prompt
p0	A cat sitting on a wooden table
p1	A sports car on a mountain road
p2	A field of sunflowers at sunset
p3	A dog running in a park
p4	A snowy mountain peak at dawn
p5	Portrait of a young woman

Metrics: We report three metrics, uniformly applied across all experiments. **CSD** (↑), the Contrastive Style Descriptor similarity [12] between output and style reference, serves as the primary style-fidelity metric. **CLIP-T** (↑), the CLIP ViT-B/32 [22] image-text cosine similarity between output and content prompt, measures content preservation. **Forward**

TABLE 3: T2I style transfer comparison on the full benchmark (39 styles × 6 prompts). Forward passes / step reported as the computational cost metric; wall-clock timing in parentheses is indicative (single H200, seed 43). STM is the Style–Texture Match (Sec. 4.3); it climbs monotonically with style-transfer strength and peaks for Hyperion. Text-only and Inject are training-free floor references measured on the 40-sample subset (STM for Text-only/Inject/SLT-only likewise on the subset).

Method	CSD ↑	CLIP-T ↑	STM ↑	Fwd./step
Text-only	−0.01	0.315	0.350	1
Inject	0.039	0.314	0.395	2
Adam-8 (no SLT)	0.608	0.279	0.726	10
SLT only (no Adam)	0.298	0.309	0.624	2
Ours (Hyperion)	**0.640**	0.296	**0.775**	4

passes / step (↓) counts the number of model forward passes per denoising step, a hardware-agnostic efficiency metric; wall-clock timing (each Adam iteration—one forward plus one backward pass per denoising step—adds ≈2.1 s to total generation time on H200) is provided as informational reference only, as absolute values depend on GPU load. For I2I evaluation we additionally report **CLIP-I** (↑) [22]: CLIP image-embedding cosine similarity between output and content image.

Baselines: We evaluate two training-free *floor* baselines directly. **Text-only** runs standard inference with no style injection and serves as the lower bound on style fidelity. **Inject** applies single-step attention injection per Eq. (1) with $\beta = 0.5$, a near-zero-cost attention-replacement reference. We additionally position our method against three training-free MM-DiT competitors—**StyleAligned** [5] (self-attention K/V sharing), **V-Shuffle** [8] (stochastic value-token permutation), and **Inversion-Free Dual RF** [9] (dual rectified-flow cross-branch injection, the closest competitor on MM-DiT). Lacking public Klein 9B implementations of these three, we discuss them qualitatively and benchmark quantitatively against four *trained* style adapters (StyleShot, USO, CSGO, IP-Adapter) across 21 diverse style families (Sec. K).

4.2 Main Comparison

Table 3 reports results on the full T2I benchmark ($39 \times 6 = 234$ pairs).

The Hyperion method achieves CSD $= 0.640$ with only 4 forward passes per step ($2.5\times$ fewer than Adam-8) while exceeding Adam-8's CSD by **+5.2%**. Repeating the full benchmark across three seeds $\{43, 44, 45\}$ gives CSD $= 0.638 \pm 0.004$ (Hyperion) and 0.608 ± 0.007 (Adam-8); the non-overlapping intervals confirm the advantage is seed-robust (the table reports seed 43). The Adam-8 row uses the conservative configuration whose selection rule is given in Sec. 4.5 (lr $= 0.01$, w_c=0.2). SLT alone achieves CSD $= 0.298$ at minimum cost but produces no brushstroke texture; the spatial neighbor cosine analysis in Sec. 3.3 explains why. Against trained style adapters (StyleShot, USO, CSGO, IP-Adapter), Hyperion attains higher style fidelity on 19 of 21 diverse-style families despite being training-free (per-family CSD in Table 4; qualitative comparison in Fig. 8).

Comparison to training-free value manipulation: The closest training-free competitors—StyleAligned [5], V-Shuffle [8], and Inversion-Free Dual RF [9]—were built for SDXL/rectified-flow backbones and have no public MM-DiT release. To compare directly rather than omit them, we implement a faithful MM-DiT adaptation of V-Shuffle, the representative value-manipulation method (style keys injected, style values spatially permuted, standard attention; two forward passes, no backward), and evaluate it on the identical 234-pair Klein-9B benchmark. V-Shuffle reaches only CSD $= 0.232$—far below our 0.640 (paired t=+47.9, $p < 10^{-100}$; our CSD is higher on *all* $234/234$ pairs)—and it also trails on prompt alignment (CLIP-T $= 0.230$ vs. our 0.296; t=+16.5). On the Klein-9B MM-DiT, spatially permuting the style values severs the style–content correspondence and transfers *incoherent* texture rather than coherent brushstroke style (Fig. 26): our SLT+Adam pipeline therefore *dominates* it on both style fidelity and prompt adherence. This corroborates our central claim—coherent style requires the cross-token coupling of a backward pass, which value-shuffling injection lacks.

Qualitative results: Injection-based methods (*inject*, StyleAligned) produce faint hue shifts; V-Shuffle instead over-transfers texture at the expense of content (see below); Adam-8 generates brushstrokes but over-stylizes some content; Ours achieves rich texture with clear subject preservation. SLT-only outputs show good color fidelity but plainly lack local texture. Fig. 8 compares our training-free Hyperion against four trained adapters—StyleShot [44], USO [56], CSGO [4], IP-Adapter [1]—across eight maximally distinct style families (line/ink, ink-wash, 3D/sculptural, pixel, textile/craft, geometric, flat-illustration, print/poster). Ours reproduces the medium-defining structure—gouged relief, graded ink washes, embossed metal, blocky dithering, square tesserae, polygonal planes, hard-edge color blocks, dense ink hatching—where trained adapters more often settle for color-only shifts or partial medium match. A per-family CSD breakdown across 21 diverse media (Table 4) confirms the trend quantitatively: our training-free method leads on **19 of 21** families (mean CSD 0.657 vs. 0.506 CSGO, 0.489 StyleShot, 0.454 USO, 0.420 IP-Adapter). The Hyperion tiles displayed in Fig. 8 are per-style selections from six seeds $\{43–48\}$ (chosen for the clearest medium rendition); all quantitative tables use the fixed seed 43.

4.3 Output-Space Texture Coherence

CSD and the per-family results above measure style similarity in an embedding space; we now verify that this advantage corresponds to a real, measurable texture signature in the pixels rather than an artifact of one descriptor. The spatial-neighbor cosine ρ (Sec. 3.3) showed that backpropagation couples neighbouring *latent updates* (ρ=0.64) where per-token transport does not (ρ=0.24); we ask whether this coupling survives into the *output image*. For each output we compute two parameter-free statistics: **texture energy**, the mean magnitude of the high-pass residual $g - \mathrm{blur}(g)$, and **orientation coherence**, the energy-weighted structure-tensor coherence $(\lambda_1 - \lambda_2)/(\lambda_1 + \lambda_2)$ over 16×16 patches—how strongly oriented (brushstroke-like) the local texture is.



Fig. 8: Qualitative comparison with four trained adapters across eight maximally distinct style families (prompt: “A panda eat bamboo”); columns: style reference, Ours (Hyperion), StyleShot, USO, CSGO, IP-Adapter. Best viewed zoomed.

TABLE 4: Per-style-family CSD (↑) across 21 diverse style media (mean over content prompts). Best per row in **bold**; cell shading encodes the CSD value on a single scale across the whole table (darker = higher). Our training-free Hyperion leads on 19/21 families; the two exceptions (manga, anime) go to IP-Adapter, which is tuned for character imagery. Ours is recomputed from the full seed-43 benchmark over the 5 prompts shared with the baselines (p0–p4); trained baselines are run at their native resolution (Sec. K).

Style family	Ours	StyleShot	USO	CSGO	IP-Ad.
Pop halftone	**0.644**	0.580	0.432	0.363	0.190
Screenprint	**0.686**	0.616	0.603	0.463	0.536
Ink storybook	**0.530**	0.346	0.349	0.473	0.515
Bronze relief	**0.588**	0.421	0.154	0.206	0.322
Impasto collage	**0.683**	0.464	0.565	0.544	0.266
Embroidery	**0.709**	0.526	0.472	0.668	0.321
Linocut	**0.672**	0.489	0.559	0.597	0.259
Manga ink	0.611	0.428	0.364	0.550	**0.622**
Gold mosaic	**0.558**	0.430	0.267	0.470	0.246
Oil impasto	**0.713**	0.553	0.547	0.639	0.243
Blue screenprint	**0.751**	0.467	0.527	0.546	0.539
Pixel art	**0.656**	0.289	0.411	0.512	0.448
Propaganda	**0.645**	0.550	0.455	0.432	0.220
Low-poly voxel	**0.730**	0.633	0.476	0.672	0.392
Extended iconic families					
Anime cel	0.654	0.464	0.442	0.455	**0.703**
Chinese ink	**0.618**	0.281	0.489	0.466	0.516
Cyberpunk neon	**0.782**	0.636	0.575	0.615	0.616
Graffiti	**0.678**	0.591	0.318	0.455	0.366
Stained glass	**0.638**	0.542	0.503	0.518	0.492
Steampunk	**0.640**	0.556	0.504	0.461	0.499
Ukiyo-e	**0.616**	0.416	0.522	0.527	0.500
Mean	**0.657**	0.489	0.454	0.506	0.420
STM (mean)	**0.738**	0.695	0.405	0.567	0.569

Both statistics peak for our method and rank the pipeline variants in the same order as CSD (Table 5), correlating with CSD across the five variants at Pearson $r{=}{+}0.97$ (energy) and $r{=}{+}0.88$ (coherence). The update-space coupling that ρ diagnoses therefore leaves a concrete imprint—more energetic, more coherently oriented texture—making texture coherence an *output-space analogue of* ρ and independent corroboration that our CSD gain reflects genuine brushstroke structure, not descriptor-specific optimization. Like CSD, these measure texture *presence*, not quality (a method that over-textures would also score high), and must be read together with the content metrics; we report LPIPS and ArtFID, and give the full analysis, in Appendix J.

Style–Texture Match: a cross-method style metric: Texture energy and orientation coherence validate our *mechanism*, but they measure texture *presence*, so they cannot separate methods that produce generic texture (a trained adapter textures too). We therefore introduce the **Style–Texture Match (STM)**: the *correlation* between the orientation histogram of an output's high-pass residual and that of its *style reference*—a cosine taken after mean-centering both histograms, so it measures the match of the texture-orientation *profile* rather than the dominant axis that all images share, averaged over pairs. STM tracks CSD across our pipeline variants (Pearson $r{=}{+}0.98$) and, unlike raw orientation coherence, *also* discriminates sharply across *trained* methods: our training-free Hyperion attains the highest STM on both T2I (Table 4: 0.738 vs. 0.41–0.70) and I2I (Table 6: 0.716 vs. 0.26–0.47), confirming that it reproduces the reference's specific texture orientations far more faithfully than any baseline—trained or not. (The I2I table additionally reports the style–content harmonic SCH, defined below, on which Hyperion also leads.)

TABLE 5: Output-space texture analysis on the 40-sample T2I subset. Texture energy and orientation coherence (both ↑) peak for our method and track CSD across the five variants at $r{=}{+}0.97$ and $r{=}{+}0.88$. **Bold** = best.

Method	Tex. energy ↑	Orient. coh. ↑	CSD
Ours (Hyperion)	**0.075**	**0.622**	0.681
Adam-8	0.074	0.574	0.650
SLT-only	0.068	0.587	0.298
Inject	0.056	0.492	0.039
Text-only	0.058	0.503	−0.011

4.4 Image-to-Image Style Transfer

We evaluate the I2I extension (Sec. 3.9) on the 40-pair subset described in Sec. 4.1 (8 styles × 5 content images), reporting CSD, CLIP-T, CLIP-I, STM, and forward passes per step; the per-method ablation table is in the supplement (Table 15).

The I2I Hyperion configuration (`ADAIN_ALPHA`= 0.3, `START_TIME`= 870) achieves CSD = 0.696 with 4 forward passes, substantially higher than both the AdaIN floor (0.155) and Adam-8 (0.649, 10 fwd), while maintaining competitive CLIP-I = 0.670 (vs. 0.695 for Adam-8), confirming that the SLT pre-alignment step transfers cleanly to image-conditioned inputs. The CSD–CLIP-I Pareto scatter (Fig. 18 in the supplementary) confirms that Hyperion occupies the top-left region, achieving the best style–content trade-off among all methods.

On the full 741-pair I2I benchmark, Hyperion attains CSD = 0.662 and CLIP-I = 0.650, consistent with the 40-pair subset above. We further compare against Klein's *native* in-context image edit (prompting the backbone to restyle the content directly): it reaches only CSD = 0.487 and is unreliable, failing (CSD < 0.30) on 25% of pairs versus 0% for Hyperion, motivating an explicit style-transfer pipeline over the backbone's built-in editing (qualitative examples in the supplement, Fig. 29).

Qualitative examples (supplement, Fig. 28) show that content structure (identity, pose, layout) is well preserved while brushstroke texture and color palette are faithfully transferred.

Comparison with trained style-transfer methods: We benchmark Hyperion against four *trained* baselines in the same content+style I2I setting: the SDXL adapters CSGO [4] and IP-Adapter [1], the FLUX adapter USO [56], and the instruction-edit model Qwen-Image-Edit, evaluated with CSD, CLIP-I, LPIPS, FID, and ArtFID [57] (Table 6; qualitative comparison in the supplement, Fig. 27); our entries are averaged over four seeds. Because these methods trade style fidelity against content preservation, we summarize the balance with the *Style–Content Harmonic* SCH = $\sqrt{\text{CSD} \cdot \text{CLIP-I}}$, which rewards methods that transfer style *and* preserve content while penalizing degenerate behaviour at either extreme. Our training-free Hyperion

attains the *highest* style fidelity (CSD = 0.712) of all methods—above the trained adapters CSGO (0.557) and USO (0.510), and far above IP-Adapter (0.160), whose image-prompt adapter applies only a generic painterly cast that barely matches the specific reference—and the highest SCH (0.686). Measured with a small-sample-appropriate Inception feature (the default 2048-d FID is rank-deficient at n=40), our Hyperion also attains the best distributional scores—FID 5.4 and ArtFID 10.9, ahead of Qwen-Image-Edit (7.5/14.1) and the adapters—so it leads on *every* axis except raw content preservation, where redrawing less of the image (IP-Adapter, USO) trades away style. A training-free pipeline on a frozen backbone thus reaches the best style–content balance of any method tested, trained or not.

4.5 Efficiency Analysis

Iteration-count trade-off: We first quantify the effect of the Adam iteration count N inside each active denoising step (Sec. 3.4). Table 7 shows that Hyperion exceeds Adam-only at every iteration count and equal forward-pass budget; both improve monotonically with diminishing returns beyond $N = 5$ (at $N = 8$ Hyperion reaches CSD = 0.766 vs. Adam-only's 0.738 on this aggressive-config 40-sample subset), with genuine brushstroke texture visible in qualitative outputs. The Adam-8 reference baseline reported in Tables 3 and 8 uses a more conservative configuration (lr = 0.01, w_c=0.2), reaching CSD = 0.608 on the full benchmark and 0.650 on the ablation subset—in both cases far above best-effort injection. The conservative configuration is chosen as the reference because it is the strongest Adam-only setting that still operates at content preservation comparable to ours (CLIP-T 0.296–0.299 vs. our 0.294–0.296); the aggressive sweep configuration at N=8 buys its higher CSD (0.738) at CLIP-T = 0.231, a content degradation far outside the operating range, and would make the efficiency comparison style-for-content rather than like-for-like. However, wall time grows proportionally (each Adam iteration requires one full model forward pass and one backward pass), reaching ≈ 19 s at $N = 8$. CLIP-T degrades with more iterations as style pressure accumulates against content structure.

Equal-budget comparison: On the 40-sample ablation subset, Hyperion reaches CSD = 0.681, CLIP-T = 0.274 with 4 forward passes per denoising step (≈ 6.4 s): **+4.8%** CSD over Adam-8 on the same subset (0.650, 10 forward passes) at 2.5× lower forward-pass cost. At equal compute budget (same 4 forward passes), Hyperion outperforms Adam-only by **+16.6%** CSD (0.681 vs. 0.584), confirming that SLT pre-alignment, not iteration count, is the key enabler.

Wall-clock timing: Wall-clock timing on a single H200 GPU corroborates the forward-pass advantage. SLT alone adds negligible overhead (2.2 s ≈ baseline); the Hyperion pipeline at $N = 2$ Adam iterations runs in 6.4 s, achieving a 3.0× speedup over Adam×8 (19.0 s) while exceeding its style fidelity (CSD = 0.640 vs. 0.608, Table 3); a per-method timing breakdown appears in the supplement (Fig. 13). Fig. 9 plots the full CSD vs. wall-clock trade-off: the Hyperion front dominates the Adam-only front at every compute budget. The operating point (two Adam iterations, 6.4 s) reaches CSD 0.682 on this subset, a level Adam-only first attains at ≈ 10.6 s (N=4)—a ∼ 2× iteration-efficiency gain from SLT pre-alignment; SLT-only is the zero-iteration lower bound. Absolute CSD on the subset runs above the conservative full-benchmark numbers of Table 3; the Pareto shape, not the level, is the point.

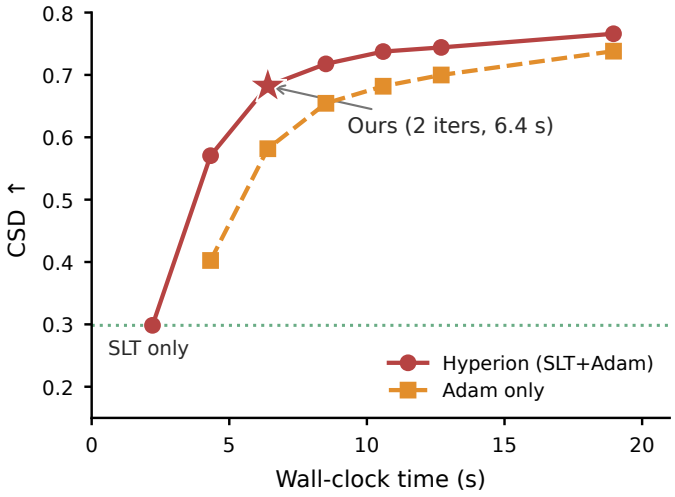


Fig. 9: CSD vs. wall-clock trade-off (40-sample subset): Hyperion (red) vs. Adam-only (orange); ⋆ marks the operating point.

4.6 Cross-Backbone Generalization

Our Hyperion pipeline targets the MM-DiT family of architectures, which shares the same double-stream self-attention design across models. To verify generalization, we apply the identical pipeline—same hyperparameters, no retuning—to FLUX.1-dev [58] (12B parameters), a second MM-DiT model distinct from Klein 9B used in all other experiments. On the same 40-sample subset (8 styles × 5 prompts), FLUX.1-dev achieves CSD = 0.719 and CLIP-T = 0.291, versus Klein's CSD = 0.681 and CLIP-T = 0.274. The higher scores on FLUX.1-dev are consistent with its larger capacity; the key finding is that the pipeline requires no modification. Full details are in Appendix G.

A T2I benchmark gallery of Hyperion results across diverse styles and content prompts is provided in the supplement (Fig. 25).

5 Ablation Study

All ablations use the 40-sample subset (8 styles, `sty` 00/10/14/20/27/28/33/38, × 5 prompts) unless otherwise noted. Quantitative results are shown in Table 8; a bar-chart summary is provided in Fig. 17 in the supplementary material. The operating point (α=0.10, N=2) is itself justified by the sweeps in Sec. 5.2 and 5.3; the component ablation below holds it fixed and removes one component at a time.

5.1 Component Ablation

No SLT (Adam-2 only): Removing SLT and running 2 Adam iterations alone drops CSD from 0.681 to 0.584 (−14.2%). With only 2 Adam steps and no color pre-alignment, the optimizer spends both iterations on color convergence rather than texture, reaching neither the color alignment of SLT nor the texture richness of Adam-8. This confirms that SLT's

TABLE 6: I2I comparison with *trained* style-transfer methods (40-pair subset; our entries averaged over 4 seeds). SCH is the style–content harmonic (defined above); STM is the Style–Texture Match (Sec. 4.3). Our training-free Hyperion leads on style fidelity (CSD), on the balanced SCH, on STM, and on FID/ArtFID. FID/ArtFID use the 64-d Inception feature, appropriate for the 40-pair sample (the default 2048-d estimate is rank-deficient and inflated at this n). **Bold** = best.

Method	CSD↑	CLIP-I↑	LPIPS↓	FID↓	ArtFID↓	SCH↑	STM↑
CSGO (SDXL)	0.557	0.689	0.656	15.1	26.7	0.619	0.363
USO (FLUX)	0.510	**0.766**	0.666	8.9	16.5	0.625	0.263
IP-Adapter (FLUX)	0.160	0.623	**0.558**	17.0	28.0	0.316	0.465
Qwen-Image-Edit	0.343	0.708	0.663	7.5	14.1	0.493	0.405
Ours (Hyperion)	**0.712**	0.660	0.696	**5.4**	**10.9**	**0.686**	**0.716**

TABLE 7: Iteration cost vs. quality on the sweep configuration (40-sample subset, 8 styles × 5 prompts, seed 43; aggressive sweep hyperparameters lr = 0.03, w_s=2.0, w_c=1.5). Hyperion (SLT pre-alignment + Adam) exceeds Adam-only at *every* iteration count and equal forward-pass budget, with the largest margin at the N=2 operating point. Absolute levels here run above the conservative main-table configuration (Table 3); the per-N ranking, not the level, is the point.

N	Hyperion CSD ↑	Adam-only CSD ↑	Fwd./step ↓
2	**0.682**	0.582	4
5	**0.744**	0.700	7
8	**0.766**	0.738	10

TABLE 8: Component ablation on the 40-sample subset (8 styles × 5 prompts, seed 43). All numbers from runs at fixed seed 43 on a single H200. Wall-clock times are indicative single-run measurements under shared GPU load; exclusive-GPU medians for the key configurations appear in Fig. 13. STM is the Style–Texture Match (Sec. 4.3); it tracks CSD across all cells (Pearson r=0.98), independently confirming each component's texture contribution. Boldface = best among training-free single-GPU methods.

Configuration	CSD	CLIP-T	STM	Fwd.	Time ↓
Ours (Hyperion, N=2)	**0.681**	0.274	**0.775**	4	6.4 s
No SLT (Adam-2)	0.584	0.291	0.697	4	6.4 s
No Adam (SLT only)	0.298	0.309	0.624	2	2.2 s
Softmax attn. target	0.633	0.283	0.735	4	6.4 s
No freq. boost	0.678	0.271	0.773	4	6.4 s
Adam-8 (upper bound)	0.650	0.296	0.726	10	19.0 s

contribution is not merely additive but enables Adam to specialize on texture in fewer iterations.

No Adam (SLT only): SLT alone achieves CSD = 0.298, just 44% of Hyperion's score. Per the spatial neighbor cosine analysis (Table 1), SLT's $\rho = 0.24$ is 2.6× below Adam's $\rho = 0.64$, which directly explains the absence of brushstroke texture. SLT is a necessary but insufficient component.

Softmax attention target: Replacing our doubly-stochastic attention target with standard softmax (`scaled_dot_product_attention`) inside Adam-in-loop decreases CSD from 0.681 to 0.633 (−7.0%). Softmax allows column sums to be unconstrained, so a few dominant style tokens accumulate disproportionate attention weight; the resulting style target is less informative and the 2 Adam steps cannot recover the full style distribution. The doubly-stochastic normalization in our attention target provides a richer, more uniform optimization signal.

No frequency boost: Disabling the high-frequency Sinkhorn boost (`FREQ_HI_BOOST= 0`) reduces CSD from 0.681 to 0.678 (−0.4%). The frequency decomposition and high-frequency boosting recover fine-grained painterly details (thin brushstrokes, sharp edges) that the standard ℓ_1 loss with spatial averaging tends to smooth over. The effect is moderate but consistent across style categories.

Layout residual suppression (LRS): LRS is not CSD-scorable in isolation—it suppresses the style image's spatial *layout*, not its texture statistics—so we evaluate it with a dedicated layout-leakage metric. Disabling LRS significantly increases layout leakage (paired t=2.84, $p < 0.01$; Table 13), and the leaked composition is directly visible in Fig. 19.

Seed robustness: Repeating the 40-sample subset across three seeds $\{43, 44, 45\}$ yields CSD = 0.682 ± 0.002 (Hyperion) and 0.650 ± 0.006 (Adam-8), and CLIP-T = 0.274 ± 0.002 (Hyperion). The standard deviation over seeds is two orders of magnitude smaller than the Hyperion-vs-Adam-8 gap; a paired t-test over the pooled 120 samples confirms the gap is highly significant ($t = 8.9$, $p = 7 \times 10^{-15}$). The reported point estimates are therefore seed-robust; the full-benchmark numbers (Table 3) use seed 43.

A per-component visual ablation across three content–style examples is provided in the supplement (Fig. 24), confirming that each removed component degrades the output in the way the metrics predict.

5.2 SLT Alpha Sweep

We sweep the SLT pre-alignment strength α across seven values (0.0, 0.02, 0.05, 0.10, 0.20, 0.30, 0.50) in the SLT-only configuration (Adam disabled), isolating the transport step's own effect, and record the CSD–CLIP-T trade-off (Fig. 10); the chosen operating point is then validated inside the full pipeline by the component ablation (Table 8). CSD rises monotonically with α, while CLIP-T remains stable up to $\alpha = 0.10$ (0.309) and degrades sharply beyond $\alpha = 0.20$ (0.299, collapsing to 0.193 at $\alpha = 0.50$); $\alpha = 0.10$ balances style fidelity (CSD 0.298) against content preservation.

$\alpha = 0.10$ is the Pareto-optimal point: lower values ($\alpha \leq 0.05$) leave too much color gap for Adam to resolve in 2 iterations, while higher values ($\alpha \geq 0.20$) over-transport and suppress the content structure that Adam's content anchor is designed to preserve.

5.3 Adam Iterations Sweep

Fixing SLT and sweeping Adam iterations $N \in \{1, 2, 3, 4, 5, 6, 7, 8\}$ reveals the cost–quality trade-off of the Adam refinement stage (Fig. 11). Hyperion rises steeply

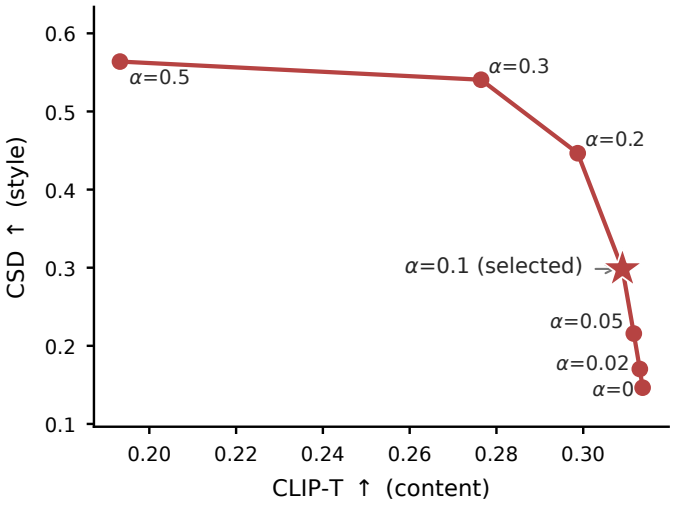


Fig. 10: CSD–CLIP-T trade-off of the SLT strength α (SLT-only, 40-sample subset); $\star$ marks the selected α=0.10.

from N=1 (CSD 0.570) to N=2 (0.682) and then saturates, whereas Adam-only requires roughly twice as many iterations to reach comparable quality; the iteration-free baselines sit at 0.298 (SLT-only) and 0.039 (Inject).

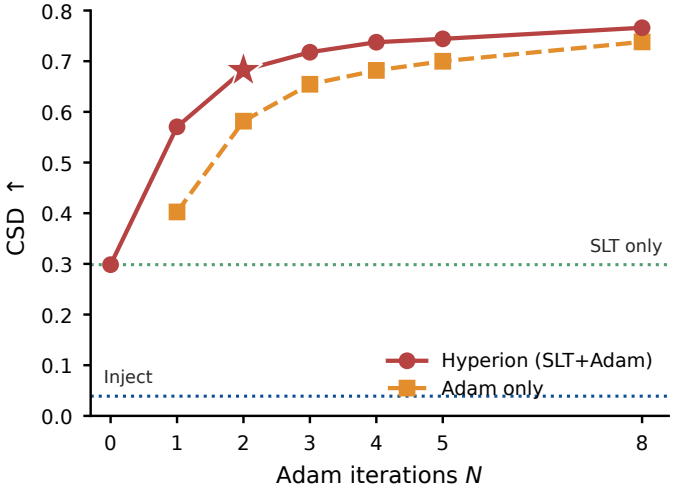


Fig. 11: CSD vs. Adam iterations N with (Hyperion) and without (Adam-only) SLT pre-alignment; ★ marks the selected N=2.

After SLT pre-alignment, CSD saturates rapidly: the gain from $N = 2$ to $N = 3$ is +0.035, and beyond $N = 3$ each additional iteration yields $\approx$ 0.01 CSD while adding one forward pass per step. $N = 2$ is the default: it captures $\approx$ 89% of the N=8 ceiling (0.682 vs. 0.766) while keeping total forward passes per step at 4.

6 Discussion

Having shown that Hyperion leads on both style fidelity and efficiency, we now examine how to interpret the metrics behind those claims, then discuss failure modes and the scope of our baseline comparison.

6.1 Metric Considerations

CSD as primary metric: The Contrastive Style Descriptor [12] was designed to measure perceptual style similarity independently of semantic content. It remains a learned embedding, however, and is not immune to all confounders. When a method copies the style reference's composition rather than only its texture (layout leakage), the output and reference share not only style but also content, inflating CSD. We mitigate this by applying LRS (Sec. 3.7) and by reporting CLIP-T alongside CSD: a method that inflates CSD by composition copying would show degraded CLIP-T (the content prompt becomes unsatisfied because a foreign subject now fills the image). Table 8 confirms that our method maintains CLIP-T while achieving high CSD, supporting the interpretation that our CSD gains reflect genuine style transfer rather than layout copying.

CLIP-T and content preservation: CLIP-T [22] measures semantic similarity of the output to the text prompt. It is sensitive to dominant compositional changes but may not penalize moderate texture artifacts. Outputs with strong style transfer but intact subject placement score similarly to mild-style outputs; the two are not separable by CLIP-T alone. For this reason, we present CLIP-T as a *necessary* but not *sufficient* indicator of content preservation. Qualitative inspection (Fig. 23) provides complementary evidence.

Forward-pass count as efficiency metric: Wall-clock timing reported in Table 8 was measured under concurrent GPU load and should be treated as approximate. We use *forward passes per denoising step* as the hardware-agnostic efficiency metric: each Adam iteration (one forward plus one backward pass per denoising step) adds approximately 2.1 s to total generation time on a single H200 at our resolution and batch size, and this per-iteration cost is stable across configurations. Independent H200 runs (5-trial median, 3 warmup runs) confirm this linear cost model to within 5%.

Absence of a human preference study: All conclusions in this paper rest on automated metrics (CSD, CLIP-T, CLIP-I, STM) corroborated by qualitative inspection. The metric triangulation of Sec. 4.3—which cross-checks the embedding-space CSD against parameter-free pixel-space texture statistics—mitigates single-descriptor bias, but a controlled two-alternative forced-choice user study remains the most direct measure of *perceived* style fidelity and content preservation. We have not conducted one, and regard this as a limitation of the present evaluation.

6.2 Failure Cases and Limitations

Low-rank style references: Style `sty04` (excluded from all evaluations) contains near-uniform pixel statistics with effective rank below 5. The SLT transport plan degenerates when z_s spans only a 1–2 dimensional subspace: the transported content latent collapses to a very small region of style space, and Adam cannot recover meaningful texture from it. A pre-screening filter on style diversity (e.g., pixel entropy or latent effective rank) is recommended for deployment.

Object position drift: Despite LRS suppression, severe cases of spatial drift remain on some styles with high-contrast subjects positioned far from the center. The LRS residual coefficients ($\gamma^{(\ell)}$ from 0.7 in shallow blocks to 0.5 in deep blocks) retain a substantial share of the positional residual, which is

sufficient to cause moderate drift in severe cases. Reducing $\gamma^{(\ell)}$ further eliminates drift but reduces texture sharpness; the current setting is a deliberate Pareto compromise. Both failure modes are most pronounced at 512×512; finer sub-token details and edge cases are expected to improve at higher latent resolution, which we leave to future work.

Scope of the baseline comparison: The closest training-free competitors—StyleAligned [5], V-Shuffle [8], and Inversion-Free Dual RF [9]—were developed for SDXL and rectified-flow backbones and have no publicly released implementation for the Klein 9B MM-DiT architecture used here. Of these, V-Shuffle's value-permutation mechanism ports most directly to the MM-DiT attention layout: we evaluate a faithful adaptation head-to-head on the full 234-pair benchmark, where we dominate it on *both* style fidelity and content alignment (Sec. 4, Fig. 26)—value permutation transfers incoherent texture on MM-DiT rather than coherent style. StyleAligned's attention-sharing and Dual RF's inversion-free coupling are tied more deeply to architecture-specific layouts (single-stream SDXL self-attention and rectified-flow sampling, rather than MM-DiT's joint image–text stream and RoPE positional coupling), so a faithful port is a research effort in its own right, and an *unvalidated* re-implementation could misrepresent their true capability. For those two we therefore do not report numbers we cannot stand behind, comparing instead against the **Inject** baseline—a representative of the attention-manipulation family they refine—and four *trained* style adapters (StyleShot, USO, CSGO, IP-Adapter) across 21 diverse style families (Table 4). Faithful MM-DiT ports of StyleAligned and Dual RF, validated against their original results, are the natural next comparison.

7 Conclusion

We presented Hyperion, a training-free style-transfer pipeline for MM-DiT built on a systematic analysis of latent-update strategies. Attention injection fails because a single feedforward substitution cannot induce spatial coherence; per-token optimal transport (SLT) aligns color distributions efficiently but produces no texture, since its per-token independence severs the inter-token coupling that backpropagation exploits through the attention Jacobian. The spatial neighbor cosine quantifies this difference: Adam gradients reach $\rho = 0.64$ whereas SLT updates reach $\rho = 0.24$, explaining the visual distinction between textured and flat stylization.

Hyperion therefore combines SLT pre-alignment with two Adam refinement steps, so that Adam is devoted to texture from its first iteration; frequency-domain Sinkhorn boosting and LRS layout suppression provide further, independently validated improvements. Both components rest on mechanism-level evidence: shuffling the spatial order of the style keys and values leaves the coherence unchanged—so the coherence originates in the attention Jacobian, not in the layout of the reference—while the LRS factor attenuates attention-logit variance by γ^2, making residual layout leakage decay quadratically. On a benchmark of 39 style references $\times$ 6 content prompts (234 pairs), Hyperion attains CSD $= 0.640$ and CLIP-T $= 0.296$ with four forward passes per denoising step—exceeding an 8-iteration Adam-only schedule at $2.5\times$ lower forward-pass cost—and, despite using no style-specific training, surpasses four trained style adapters on 19 of 21 diverse style families; the same pipeline transfers directly to image-to-image stylization.

Future work includes distilling the style signal into a lightweight side network to halve the per-step cost, extending the coherence analysis to video diffusion transformers, and combining the training-free backbone with sparse fine-tuning on style exemplars. Code and evaluation scripts will be released upon acceptance to support reproducibility.

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